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**PROJECT TITLE : RaktSanket: A Composite Blood Shortage-Response Index for
Proactive Donor Pre-Activation Using Explainable AI**

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RaktSanket: AI-based Predictive Blood Availability and Donor Response System

ABSTRACT:

India's blood transfusion system runs on a persistent, well-documented shortfall. According to the National AIDS Control Organization, India's annual blood donation rate stands at roughly 7.4 million units against a shortfall of about 2.6 million units. This is not a data problem — the India Meteorological pattern of festivals, exam seasons, and monsoon disruptions that drive demand spikes is well known, and donor registries such as e-RaktKosh already track stock levels nationally.

Existing platforms are reactive: a request is raised only after a blood group runs low, and every donor on the list is notified indiscriminately, regardless of how likely they are to respond, causing donor fatigue and wasted outreach. RaktSanket addresses this gap by introducing CBSRI, the Composite Blood Shortage-Response Index, a novel weighted metric that fuses three data streams into a single, per-district, per-blood-group weekly score: a Shortage Forecast Score (SFS) derived from time-series modelling of historical demand and seasonal/festival patterns, a Donor Response-Probability Score (DRP) computed via XGBoost from donor recency, frequency, and historical response behaviour, and a Fatigue-Adjusted Priority Score (FAPS) that down-weights donors who have been repeatedly notified without responding. The weights of these three components are derived from correlation analysis on historical shortage events rather than assumed arbitrarily, making CBSRI a research contribution in its own right.

An XGBoost model predicts CBSRI seven to fourteen days in advance with SHAP explainability, showing exactly which of the three components is driving a given shortage alert and which factors are driving an individual donor's priority rank. The system is validated against documented historical shortage episodes — including the COVID-19 first and second wave collection collapses of 2020 and 2021 and a recurring festival-season stock dip — establishing that CBSRI crosses critical thresholds in advance of the shortage becoming visible in stock records. Donor notifications are delivered through a tiered, fatigue-aware cascade, with advisories generated in the operator's and donor's preferred regional language.

I. INTRODUCTION:

Blood transfusion services form an indispensable part of emergency and elective healthcare, and unlike most medical resources, blood cannot be manufactured — it depends entirely on voluntary human donation. In India, this dependency collides with a structural shortfall: national estimates place the annual blood requirement at approximately 14-15 million units against an actual collection of 11-12 million units, and geospatial studies of blood-bank access in northern India report availability rates well below WHO and Lancet Commission benchmarks, particularly in low-income and rural districts. The consequence is not abstract — inadequate access to timely, matched blood contributes directly to preventable deaths from haemorrhage, anaemia, and complications during childbirth and trauma care.

The core technological problem is not the absence of data. The National AIDS Control Organization, the e-RaktKosh national blood bank portal, and individual blood banks all record donation, stock, and usage figures. Government open-data platforms publish aggregate statistics. Academic and industry research has independently produced donor-behaviour prediction models and blood-demand forecasting models of respectable accuracy. What is missing is the mathematical fusion of these signals into a single, explainable, forward-looking index — one that tells a blood bank not only that a shortage is statistically likely, but which specific donors to contact, in which order, and why, before the shortage becomes a clinical emergency.

RaktSanket is built to close exactly this gap. It introduces CBSRI — the Composite Blood Shortage-Response Index — a unified weekly stress-and-response score computed per district and blood group, combining a shortage-forecasting layer with an explainable, fatigue-aware donor-ranking layer. An XGBoost model trained on this composite index predicts elevated shortage risk one to two weeks before it manifests as a stock shortfall. SHAP explainability ensures every alert and every donor ranking comes with a transparent reason. A tiered notification cascade then contacts the highest-probability donors first, escalating only if there is no response, while deprioritising donors who have been notified repeatedly without responding.

a. Motivation (Problem Statement):

The motivation behind RaktSanket comes from a pattern that repeats across Indian blood banks every year: a hospital discovers a shortage only when a patient is already waiting for a transfusion, and the response is a blanket SMS or call-out to the entire donor list, most of whom will not respond, some of whom are already fatigued from being notified too often, and none of whom were contacted with any lead time. Each of the underlying signals — a dip in seasonal donation rates, a spike in expected demand around festivals or accident-prone travel periods, and a donor's own historical response pattern — is measurable and, in most cases, already recorded somewhere in the system. Yet no system today fuses them into a single forward-looking, explainable score.

Existing systems fail blood banks and patients in three specific ways. First, national portals such as e-RaktKosh expose current stock levels but provide no forecast of where a shortage is likely to emerge next week. Second, blood-donor mobile apps and SMS-based systems can notify donors once a shortage is already declared, but treat this notification step as reactive and largely undifferentiated, rarely ranking donors by actual likelihood to respond and almost never explaining why a particular donor was contacted. Third, published donor-ranking and demand-forecasting research — reviewed in detail in Section III — each solves one half of the problem well, but no reviewed system combines shortage forecasting with an explainable, fatigue-aware donor-activation layer calibrated for Indian blood banks.

The specific gap RaktSanket targets is this unmeasured, un-unified lead time between a forecastable shortage risk and the fatigue-aware, explainable activation of the right donors. Quantifying this relationship as CBSRI, predicting it days in advance, and using it to drive a transparent, tiered donor-notification cascade is the precise motivation behind this project. The goal is not another blood-bank dashboard — dozens already exist — but a measurable reduction in the lag between a forecastable shortage and an effective, non-fatiguing donor response.

II. Feasibility Study:

The feasibility study for RaktSanket was conducted across three dimensions — technical, operational, and economic — following the same evaluation structure used for comparable data-driven early-warning systems in the Indian public-data ecosystem.

Technical Feasibility

RaktSanket uses publicly available data from Kaggle, NACO, e-RaktKosh, and the Open Government Data Platform to build and validate its prediction models. Its modular data pipeline allows donor datasets to be easily updated or replaced without changing the overall system architecture. Blood shortage forecasting is performed using time-series analysis and an XGBoost model, while SHAP provides explainable predictions by identifying the factors contributing to shortage risk and donor ranking. The system is developed entirely with open-source technologies, including Python, Pandas, Scikit-learn, XGBoost, SHAP, Statsmodels/Prophet, FastAPI/Flask, and React, with a free-tier LLM API used to generate multilingual donor and operator advisories.

Operational Feasibility

The hospital/blood-bank-facing dashboard is designed to run in any standard browser, requiring no specialised installation. The platform is web-based and accessible through any standard browser, requiring no specialized hardware or software installation. Donors receive automated SMS and app notifications in their preferred regional language through a fatigue-aware notification cascade, reducing unnecessary alerts while minimizing manual intervention by blood bank staff.

Economic Feasibility

The entire data and deployment stack of RaktSanket is built on zero or near-zero cost sources. Government portals (NACO, data.gov.in, e-RaktKosh) and open dataset repositories such as Kaggle are free to access. Development tools are open-source. The total estimated development cost is limited to internet charges and development hardware already available to the team, making RaktSanket both technically feasible and economically viable for academic development and eventual pilot adoption by a district blood bank or NGO partner.

III. Literature Survey / Related Work:

Existing research relevant to RaktSanket spans five domains: donor behaviour, motivation and classification prediction; explainable machine learning for donor-level decisions; blood demand and shortage forecasting; notification, outreach automation and donor engagement systems; and the methodological foundations of composite index construction and model explainability. Each area is reviewed below with the gaps that RaktSanket directly addresses.

A. Donor Behaviour, Motivation and Classification Prediction

Kauten, Gupta, Qin, and Richey [3] used operational data from a large US regional blood centre paired with Random Forest and Gradient Boosting classifiers to model donor retention, achieving a Matthews Correlation Coefficient of 0.851 and identifying scheduling-related features as the strongest predictors of return donation. Yeh, Yang, and Ting [4] introduced the Recency-Frequency-Monetary (RFM) framework for characterising blood donors using a Bernoulli sequence model, establishing the feature schema — recency, frequency, monetary volume, and time since first donation — that has since become the standard input representation for donor-prediction research worldwide. A 2022 Nature Scientific Reports study [5] built machine-learning models using thirteen donor features to rank willingness-to-donate as a floating score between 0 and 1, validating the approach in real recruitment drives during two COVID-19 outbreaks. Alajrami et al. [6] applied an artificial neural network to predict individual donation behaviour, reporting the highest reported accuracy (99.31%) on a standard donor-behaviour benchmark at the time of publication. Salazar-Concha and Ramírez-Correa [7] used a decision-tree approach to predict the intention to repeat blood donation from a limited attribute set, while Cloutier et al. [8] applied random forest models to sociodemographic, marketing, and adverse-event data to predict donation-return rates specifically among young donors. Boonyanusith and Jittamai [10], Darwiche et al. [11], Ramachandran, Girija, and Bhuvaneswari [12], and Testik et al. [13] each applied neural network, SVM, decision-tree, and clustering methods respectively to donor classification and arrival-pattern discovery, with Ramachandran et al.'s work notably conducted on Indian Red Cross Society blood bank data — one of the few studies in this domain grounded in an Indian context. Across all these studies, donor-response and donor-retention prediction is established as a mature, well-validated machine-

learning task; the consistent gap is that none of them forecasts an upcoming shortage or connects the donor-ranking output to any forward-looking demand signal.

B. Explainable Machine Learning for Donor-Level Decisions

Vinkenoog, van Leeuwen, and Janssen [9] applied explainable machine-learning models to predict haemoglobin-deferral risk in blood donors, explicitly analysing the operational consequences of model interpretability for blood-supply decision-making — one of the few studies to treat explainability as a first-class requirement rather than an afterthought in blood-donor modelling. This directly supports RaktSanket's design choice to make SHAP explainability a core system component rather than a post-hoc addition, ensuring that both the shortage-forecast component and the donor-ranking component of CBSRI expose their reasoning to the hospital operator.

C. Blood Demand and Shortage Forecasting

A 2025 study on optimizing blood transfusions in resource-constrained areas [14] combined heuristic donor-matching with a shortage-prediction module forecasting the acceptance ratio of blood requests up to ten days in advance using LSTM, ARIMA, and Linear Regression — the closest published system in scope to RaktSanket's Shortage Forecasting component. Ben Elmir, Hemmak, and Senouci [15] proposed a three-module smart platform combining blood-demand forecasting, donor classification, and appointment scheduling, piloted at a blood donation centre in Algiers; comparing ARMA, ARIMA, AutoReg, ANN, Linear Regression, and SVR, their deployed system increased collected blood volume by 11% and reduced inventory wastage by 20% over a one-year period, providing strong evidence that forecast-driven donor targeting produces measurable operational gains. Shih and Rajendran [16] compared seven time-series methods against artificial neural networks and multiple regression for forecasting the Taiwan Blood Services Foundation's blood supply, finding that seasonal ARIMA and Seasonal Exponential Smoothing produced the lowest forecast error — evidence that simpler seasonal time-series models can outperform machine-learning approaches for this task, which RaktSanket treats as a benchmark rather than assuming more complex models are automatically superior. Fortsch and Khapalova [17] evaluated multiple practical forecasting methods, including naive, exponential smoothing, moving average, and ARMA

models, on five years of daily blood-centre demand data, finding ARMA models performed best in the majority of tested settings. Lowalekar and Ravi [18], notably working with an Indian blood bank case study, applied a Theory-of-Constraints thinking-process framework to donor-arrival prediction, identifying donor age, appointment lead time, and number of prior donations among the most influential factors — directly relevant to RaktSanket's Indian deployment context. Fanoodi, Malmir, and Jahantigh [19] compared ANN and ARIMA models specifically for reducing platelet-demand uncertainty, and Twumasi and Twumasi [21] examined machine-learning approaches for forecasting blood demand from hospital data containing missing values and outliers — a practical concern directly relevant to working with incomplete Indian blood-bank records. Across all these studies, forecasting accuracy is treated as the end goal; none of them link the forecast output to an explainable, individually-ranked donor-activation mechanism.

D. Notification, Outreach Automation and Donor Engagement Systems

Chinnaswamy et al. [22] describe an intelligent system for automating blood-donor classification and notification, using SMS technology to notify different tiers of eligible donors depending on the severity of a blood-stock deficit — a tiered notification mechanism that is conceptually close to RaktSanket's own cascade, but without any predictive shortage-forecasting layer or explainability, and without any mechanism to avoid repeatedly targeting the same non-responsive donors. Müller and Reuter-Oppermann [24] applied design-science research methodology to persuasive blood-donation app and chatbot design across individualist (South Africa) and collectivist (Ghana) donation cultures, finding that the effectiveness of specific engagement features (reminders, rewards, social/normative cues) depends strongly on the target population's donation culture — a finding directly relevant to designing RaktSanket's regional-language advisory tone for Indian donor populations, which are themselves culturally heterogeneous across states. A 2025 narrative review in *Vox Sanguinis* [23] surveys the broader landscape of AI in blood donor management, covering robotic process automation, machine-learning-based donor behaviour prediction, deferral-risk prediction, personalised outreach, and AI chatbots, and documents real deployments — including the Ben Elmir et al. system's 11% collection increase and 20% wastage reduction, and a Hamilton, Ontario hybrid ML/optimisation model that reduced red blood cell inventory by 38% and costs by 43%

without increasing shortages — while also flagging open challenges of data quality, ethical use, and infrastructure that RaktSanket's methodology directly addresses through its documented, substitutable public-dataset strategy.

E. Composite Index Construction and Explainability Methodology

Lundberg and Lee [25] introduced SHapley Additive exPlanations (SHAP), the game-theoretic feature-attribution framework this project uses to make both the shortage-forecast and donor-ranking components of CBSRI individually explainable, and which has since become the standard explainability method across the donor-prediction and demand-forecasting literature reviewed above. The OECD/Joint Research Centre Handbook on Constructing Composite Indicators [26] provides the standard methodology for deriving sub-index weights through statistical correlation analysis rather than arbitrary assignment — the same weighting philosophy RaktSanket applies when deriving the relative contribution of the Shortage Forecast Score, Donor Response-Probability Score, and Fatigue-Adjusted Priority Score within CBSRI.

Identified Research Gap:

Across the studies reviewed above, no single system fuses shortage-risk forecasting, explainable donor-response ranking, and fatigue-aware notification sequencing into one composite, correlation-weighted index calibrated for Indian blood banks. Donor-response prediction (Section A) is a mature, well-validated task on its own; shortage/demand forecasting (Section C) is similarly mature; but no reviewed study combines them, and no reviewed notification system (Section D) is both explainable and fatigue-aware. No study grounds a regional-language advisory system directly in a live, explainable, composite machine-learning output rather than static templates. RaktSanket addresses each of these gaps through CBSRI, a data-driven XGBoost prediction pipeline with SHAP explainability, validation against documented historical shortage episodes, and a tiered, fatigue-aware, regional-language donor-activation cascade for Indian blood banks.

IV. Objective and Scope of the Project:

Objectives

1. To design and formulate CBSRI (Composite Blood Shortage-Response Index), a novel weighted metric combining a Shortage Forecast Score, a Donor Response-Probability Score, and a Fatigue-Adjusted Priority Score, with weights derived through correlation analysis on historical shortage-event data rather than assumed arbitrarily.
2. To develop a time-series Shortage Forecasting model capable of predicting elevated shortage risk seven to fourteen days in advance, per district and blood group, incorporating seasonal and festival-driven demand patterns specific to the Indian context.
3. To build an XGBoost-based Donor Response-Probability model trained on donor recency, frequency, and historical response behaviour, sourced from publicly available donor-behaviour datasets such as those hosted on Kaggle.
4. To integrate SHAP explainability into both the shortage-forecasting and donor-ranking components, so that every alert and every donor's priority rank comes with a transparent, human-readable reason.
5. To design a tiered, fatigue-aware notification cascade that escalates through donor tiers and deprioritises donors who have been notified repeatedly without responding.
6. To validate CBSRI against documented historical shortage episodes — including the COVID-19 first and second wave collection disruptions and recurring festival-season stock dips — benchmarking forecast quality against a naive baseline such as a 7-day moving average.
7. To deliver regional-language advisories to donors and hospital operators, grounded in real-time CBSRI scores rather than generic recommendations, through a lightweight, low-bandwidth-friendly web dashboard.

Scope

The pilot implementation targets a small set of representative districts/blood banks (to be finalised with a partner blood bank or NGO during implementation) rather than a nationwide rollout, allowing the forecasting and ranking components to be properly calibrated and validated before wider deployment. The system uses only publicly available or freely licensable data sources — donor-behaviour datasets sourced from repositories such as Kaggle, and aggregate shortage/stock statistics from NACO, data.gov.in, and e-RaktKosh — with the specific donor dataset intentionally left substitutable rather than fixed, since better or more India-specific public datasets may become available during the project timeline.

The machine-learning pipeline is limited to XGBoost with SHAP explainability for the donor-ranking component and classical/seasonal time-series methods (ARIMA/SARIMA/Prophet) for the shortage-forecasting component, benchmarked where feasible against LSTM, chosen for a balance of predictive accuracy, interpretability, and computational efficiency suitable for a lightweight, low-cost deployment. The donor- and operator-facing interface is built as a lightweight web dashboard compatible with low-bandwidth connectivity.

Aadhaar/eKYC integration, wearable integration, drone-based delivery, kidney/organ paired-exchange matching, and a full multi-platform build (separate native apps per OS) are documented as potential future extensions but are excluded from the current implementation scope, consistent with keeping the contribution narrow, well-evaluated, and completable within a single academic term

V. REQUIREMENT FOR THE PROPOSED WORK:

Software Requirements

Category	Tool / Technology
Programming Language	Python 3.x, JavaScript/TypeScript
Machine Learning	XGBoost, Scikit-learn
Time-Series Forecasting	Statsmodels (ARIMA/SARIMA), Prophet
Explainable AI	SHAP
Data Processing	Pandas, NumPy
Public Data Sources	Kaggle, data.gov.in, e-RaktKosh, NACO reports
Backend Framework	FastAPI / Flask
Frontend	React / HTML5, CSS3, JavaScript
Mapping and Visualisation	Leaflet.js, Matplotlib / Chart.js
Large Language Model	Free-tier LLM API for regional-language advisories
Database	PostgreSQL / Supabase / SQLite
Version Control	Git, GitHub
Deployment	Render / Vercel (free tier)
Operating System	Windows / Linux / macOS

Hardware Requirements:

Category	Specification
Development Machine	Minimum 8 GB RAM, Intel i5 or equivalent processor
Storage	Minimum 10 GB free disk space for datasets and model files
Internet Connectivity	Required for dataset download and API access
Deployment Server	Cloud-based free tier — sufficient for pilot scale
End-User Device	Any smartphone with SMS or basic internet connectivity

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